2.5.3 Decentralized identity: shifting control to users

Decentralized identity systems allow users to control and manage their own identity without relying on a central authority. You'll also sometimes hear this called *self-sovereign identity (SSI)*.

In the decentralized identity model, people hold credentials in a cryptographic wallet and can share them with others when they need to prove something about themselves. This is the model we use most often in the physical world. You don't have an account at the local grocery store that you must log into when you need to prove your age to buy alcohol. Instead, you show them a government-issued ID that has your birthdate on it. This works for three reasons:

- ID is relatively ***tamper-proof.***
- The ID has some means of ***proving it was issued to you,*** like a picture.
- The ID was ***issued by someone the grocery store trusts*** to do a good job of determining people's birthdates.

Decentralized identity systems use a similar model, where trusted parties issue credentials to people who hold them in wallets and can present them to RPs who verify them cryptographically. This process is decentralized because no single organization decides what credentials exist, which credentials a person should hold, or what credentials an RP must accept. Anyone can issue a credential for any purpose, hold whatever credentials they like, and trust whatever credentials they choose.

The primary advantages of the decentralized model lie in the freedoms described in the previous paragraph. By giving participants choice, decentralized identity models solve the consent problem and give people the power to manage their privacy more completely. Organizations can share data without prior agreements. For example, the grocery store has no agreement with the state that allows it to use the state-issued ID.

Even so, there's still a role for trust frameworks. RPs want to know which credentials they can trust without creating agreements with every credential issuer. These, too, are decentralized. For example, if a university issues a